Day 13 (Nov 1): Weekly Test Revision Sheet

Lists and tuples | Dicts and sets | XLOOKUP and INDEX+MATCH | Branches, clone and pull. Fresh from memory, no notes.

1. KEY IDEAS

Weekly test result

- Python (Problem 1, tuples inside a list plus a dict): **not solved**, so Dicts and sets and Lists and tuples repeat. Excel: the IFNA vs IFERROR reasoning repeats. Git: clone vs pull and the cause of a merge commit repeat.

Python: a new problem shape (took longer, more depth)

- The weekly test asked for each student's best score from a list of (name, score) tuples. The pattern was already known; what failed was **spotting it in a new shape**.
- Before writing code, say four things: what is the **key**, what is the **value**, what happens the **first time** a key appears, and what is the **update** (add 1, add the value, or compare and replace).
- If stuck, write the skeleton you do know (the empty dict, the loop that unpacks the tuple) and say the next step aloud. Partial work earns credit; a blank earns none.
- Tuples and lists: data[0][1] = 85 raises **TypeError: 'tuple' object does not support item assignment** (the tuple slot cannot change). data[0] = ("Asha", 85) works, because the list slot can. Negative start: nums[-2:] begins at the second from the end.

Excel: lookups

- **=INDEX(B2:B5, MATCH("A103", A2:A5, 0))**: MATCH gives the position inside the range (3), INDEX reads it (Bag). The 0 means exact match.
- **XLOOKUP(lookup, lookup_array, return_array, if_not_found)**: the 4th argument is the message. A missing ID with no message gives #N/A.
- IFNA wrapped around the **whole** INDEX+MATCH catches the MATCH error, because INDEX passes #N/A through. IFERROR also hides #REF!, #VALUE!, #NULL!, #NAME?, #DIV/0! and #NUM!, which are real mistakes.

Git

- **Fast-forward**: main has no new commits since the branch was made. **Merge commit**: both branches have new commits (diverged), 2 parents, 'Merge made by the 'ort' strategy.' **Conflict**: same lines of the same file.
- **Clone** copies a repository (usually a remote one, a local folder works too) with its history and sets origin. **Pull** = fetch (download new commits) + merge (into your current branch).

2. EXAMPLES TO KNOW

```
=INDEX(B2:B5, MATCH("A103", A2:A5, 0))                  -> MATCH 3, result Bag
=XLOOKUP("A999", A2:A5, B2:B5, "Order not found")        -> Order not found
=IFNA(INDEX(B2:B5, MATCH("A999", A2:A5, 0)), "Order not found")  -> Order not found
=MATCH("A999", A2:A5, 0)                                   -> #N/A
```

```
*   d8ec229 (HEAD -> main) merge the updates     # merge commit, 2 parents
|\
| * 314974d (feature-y) Update log on feature-y  # new commit on the branch
* | a49b668 Add main information                 # new commit on main (different file)
|/
* 6234b4f Initial commit                        # common ancestor
```

3. MY MISTAKES AND THE FIX

What I said or did	The fix
Stuck 25 minutes on a tuple + dict problem in a new shape	Name the key, value, first-time step and update before coding. If stuck, write the skeleton.
IFNA 'only handles the inner MATCH error'	Wrapped around the whole formula it catches the #N/A that MATCH passes through INDEX.
Merge commit happens when 'different files change'	It happens when both branches have new commits (diverged). Files do not decide it.
Clone copies 'a local repository'	Clone copies a repository, usually remote, with history, and sets origin.
Pull 'updates the changes made in that repo'	Pull = fetch (download) + merge into the current branch.
No reason given for several answers	Always: output AND reason AND value.

4. HOW I IMPROVED

- Excel: INDEX+MATCH and XLOOKUP formulas, values and error names were right in the test.
- Git: ran the full diverged-merge sequence myself and read the graph, the 'ort' message and the 2 parents correctly.
- Time: left the stuck Python problem after 25 minutes and finished the rest, instead of losing the whole test.

5. PRACTICE TO RETRY (no solutions)

- records = [("Asha", 80), ("Ravi", 65), ("Asha", 90)]: build each student's best score. Then try the order reversed.
- Same records: build each student's total. What changes in the update step?
- Product table in A2:B5: write INDEX+MATCH wrapped in IFNA with the message "Not listed", and test one ID that exists and one that does not.
- Git, fresh folder: make a diverged merge on different files, then clone the repo to a second folder.

6. RECALL QUESTIONS (answer from memory, no notes)

- What four things do you decide before writing a dict problem?
- What error does data[0][1] = 85 raise when data is a list of tuples? Does data[0] = ("Asha", 85) work?
- Why does IFNA around the whole INDEX+MATCH catch a missing ID? What does IFERROR risk?
- What forces a merge commit? What do clone and pull each do?

7. OPEN ITEMS

- Python retake on a new problem. Business topic, AI Fluency and AI skill hour moved to Nov 3 and Nov 4.
- Dicts and sets and Lists and tuples repeat; clone vs pull needs one more check.